

Measure-Valued solutions of Euler System

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August 15, 2026

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Chapter 1

Preliminaries

We denote a general Banach space with X with norm $\|\cdot\|_X$ throughout this section. X^* denoted the dual of the space X and duality pairing between X and X^* will be given by $\langle \cdot, \cdot \rangle_{X^*, X}$. We use $\langle \cdot, \cdot \rangle$ to indicate the inner product in X .

When $X = L^p(\Omega)$, $1 \leq p \leq \infty$, we assume that $\Omega \subset \mathbb{R}^d$ is bounded and measurable. The norm is denoted by $\|\cdot\|_{L^p}$ or $\|\cdot\|_p$. Further, $|A|$ denotes the Lebesgue measure of a subset $A \subset \Omega$.

Definition 1 (Weak and Weak-* Convergence). We say that a sequence $\{x_n\}_{n \in \mathbb{N}} \subset X$ converges weakly to $x \in X$, denoted by $x_n \rightharpoonup x$, if

$$\langle x^*, x_n \rangle_{X^*, X} \rightarrow \langle x^*, x \rangle_{X^*, X} \text{ for any } x^* \in X^* \text{ as } n \rightarrow \infty.$$

We say that a sequence $\{x_n^*\}_{n \in \mathbb{N}} \subset X^*$ converges weakly-* to $x^* \in X^*$, denoted by $x_n^* \rightharpoonup^* x^*$, if

$$\langle x_n^*, x \rangle_{X^*, X} \rightarrow \langle x^*, x \rangle_{X^*, X} \text{ for any } x \in X \text{ as } n \rightarrow \infty.$$

1.1 Weak Compactness in Banach Spaces

Theorem 1 (Alaoglu's Theorem). *Let X be a Banach space. Then the unit ball of X^* , B_{X^*} , is weakly-* compact. Further, if X is separable, then the subspace topology induced on B_{X^*} via the weak-* topology is metrizable. If X is reflexive, then the unit ball of X , B_X , is weakly compact.*

Theorem 2 (Banach-Eberlein-Smulian Theorem). *Let X be a Banach space.*

- 1. X is reflexive if and only if every uniformly bounded sequence in X has a weakly convergent subsequence.*
- 2. If X is separable, then every uniformly bounded sequence in X^* has a weakly-* convergent subsequence.*

1.1.1 Weak compactness in L^p

The dual space of $L^p(\Omega)$ is the space $L^q(\Omega)$, where $\frac{1}{p} + \frac{1}{q} = 1$ for $1 \leq p < \infty$, with the duality pairing

$$\langle f, g \rangle_{q,p} = \int_{\Omega} f(y)g(y)dy, \quad f \in L^q(\Omega) \text{ and } g \in L^p(\Omega).$$

Since L^p is separable for $1 \leq p < \infty$ and reflexive for $1 < p < \infty$, we have the following characterization result for $1 < p \leq \infty$.

Lemma 1 (Weak Compactness in L^p). *For $1 < p \leq \infty$, let $\{f_n\}_{n \in \mathbb{N}} \subset L^p(\Omega)$ be uniformly bounded. Then, there exists a subsequence $\{f_{n_k}\}_{k \in \mathbb{N}}$ and $f \in L^p(\Omega)$ such that*

$$\begin{aligned} f_{n_k} &\rightharpoonup f \text{ in } L^p(\Omega), & \text{if } 1 < p < \infty, \\ f_{n_k} &\xrightarrow{*} f \text{ in } L^\infty(\Omega), & \text{if } p = \infty \end{aligned}$$

Proof. (Proof Idea): The existence of subsequence for $1 < p < \infty$ is a direct consequence of 2 since in this case L^p is reflexive. For the case when $p = \infty$, L^∞ is separable hence from 1, the weak-* topology is metrizable. The notion of compactness is same as sequential compactness, giving us the required weakly-* convergent subsequence. ■

Weak compactness in L^1

The Lemma 1 is not applicable in the case of $p = 1$ due to the lack of reflexivity of $L^1(\Omega)$. We also cannot use separability of $L^1(\Omega)$ and theorem 1 since there does not exist a predual in the case of $L^1(\Omega)$.

However, we can use the embedding $L^1(\Omega) \hookrightarrow \mathcal{M}(\bar{\Omega})$, where $\mathcal{M}(\bar{\Omega})$ denotes the Banach space of finite Radon measures on Ω identified as the dual of $C(\bar{\Omega})$ and have the following result:

Lemma 2. *Let $\{f_n\}_{n \in \mathbb{N}} \subset L^1(\Omega)$ be uniformly bounded. Then, there exists $\mu \in \mathcal{M}(\bar{\Omega})$ and a subsequence $\{f_{n_k}\}_{k \in \mathbb{N}}$ such that*

$$f_{n_k} \xrightarrow{*} \mu \text{ in } \mathcal{M}(\bar{\Omega}).$$

Remark 1. Given $\{\mu_n\}_{n \in \mathbb{N}} \subset \mathcal{M}(\bar{\Omega})$, we say that

$$\mu_n \xrightarrow{*} \mu \text{ in } \mathcal{M}(\bar{\Omega}), \text{ if } \int_{\Omega} \phi d\mu_n \rightarrow \int_{\Omega} \phi d\mu \text{ for all } \phi \in C(\bar{\Omega}).$$

Lemma 2 provides the weak convergence of a uniformly bounded sequence of functions belonging to $L^1(\Omega)$ to a limit in $\mathcal{M}(\bar{\Omega})$. For the weak convergence of the sequence to a limit in L^1 , we need the notion of equiintegrability which is defined as follows:

Definition 2 (Equiintegrability). A family of functions $\mathcal{F} \subset L^1(\Omega)$ is said to be equiintegrable if for every $\varepsilon > 0$, there exists $\delta_\varepsilon > 0$ such that

$$\sup_{f \in \mathcal{F}} \int_A |f| dy < \varepsilon \text{ whenever } A \subset \Omega \text{ is measurable and } |A| < \delta_\varepsilon.$$

If a sequence of $L^1(\Omega)$ functions is equiintegrable, then the sequence does not develop concentrations (means sequence does not ‘blow up’ locally).

Complete characterization of weak convergence of sequences in $L^1(\Omega)$:

Lemma 3 (Weak Convergence in L^1). *Let $\{f_n\}_{n \in \mathbb{N}} \subset L^1(\Omega)$. The following are equivalent:*

1. $\{f_n\}_{n \in \mathbb{N}}$ has a weakly convergent subsequence;
2. $\{f_n\}_{n \in \mathbb{N}}$ is uniformly bounded and equiintegrable;
3. (Dunford-Pettis criterion) for any $\varepsilon > 0$, there exists $k \in \mathbb{N}$ such that

$$\sup_{n \in \mathbb{N}} \int_{\{|f_n| \geq k\}} |f_n| dy < \varepsilon;$$

4. (De La Vallé Poussin's Criterion) there exists a non-negative function $\psi : [0, \infty] \rightarrow \mathbb{R}$ such that

(a) ψ is superlinear, i.e.

$$\lim_{\lambda \rightarrow \infty} \frac{\psi(\lambda)}{\lambda} = \infty,$$

(b) the condition

$$\sup_{n \in \mathbb{N}} \int_{\Omega} \psi(|f_n|) dy < \infty$$

holds.

1.2 Measureability of Banach-valued functions

Let X be a Banach space and $f : \Omega \rightarrow X$ be a Banach-valued function. The points of Ω will be denoted by y and elements of X will be denoted by x .

Definition 3 (Simple Function). A function $s : \Omega \rightarrow X$ is said to be a simple function if there exists measurable subsets $\Omega_1, \dots, \Omega_k \subset \Omega$ and elements $x_1, \dots, x_k \in X$ such that

$$s(y) = \sum_{i=1}^k \chi_{\Omega_i}(y) x_i,$$

where χ_{Ω_i} denotes the characteristic function of Ω_i . The integral of a simple function is defined as

$$\int_{\Omega} s(y) dy = \sum_{i=1}^k |\Omega_i| x_i.$$

Definition 4 (Strong Measurable/Bochner Measurable). f is said to be strongly measurable if there exists a sequence of simple functions $\{f_n\}_{n=1}^{\infty}$ such that for each $y \in \Omega$,

$$\|f_n(y) - f(y)\|_X \rightarrow 0 \quad \text{as } n \rightarrow \infty \quad \text{a.e. } y \in \Omega.$$

Definition 5 (Weakly measurable). f is said to be weakly measurable if the map $y \mapsto \langle x^*, f(y) \rangle_{X^*, X}$ is Lebesgue measurable for each $x^* \in X^*$.

Definition 6 (Weakly-* measurable). Let X^* be the dual space of X . A map $f : \Omega \rightarrow X^*$ is weakly-* measurable if the map $y \mapsto \langle f(y), x \rangle_{X^*, X}$ is Lebesgue measurable for each $x \in X$.

Remark 2. Let $(X, \mathcal{A})$ and $(Y, \mathcal{B})$ be measurable spaces and a map $f : X \rightarrow Y$ is $(\mathcal{A}, \mathcal{B})$ -measurable if $f^{-1}(B) \in \mathcal{A}$ for all $B \in \mathcal{B}$.

1.2.1 Function Spaces involving Banach Space-Valued Maps

Definition 7 (Space of Weakly continuous functions ($C_{weak}(\Omega; X)$)). $f : \Omega \rightarrow X$ is weakly continuous if:

- the map $y \mapsto \|f(y)\|_X$ is bounded,
- for each $x^* \in X^*$, the map $y \mapsto \langle x^*, f(y) \rangle_{X^*, X}$ is continuous.

The norm is defined as

$$\|f\|_{C_{weak}(\Omega; X)} = \sup_{y \in \Omega} \|f(y)\|_X$$

Definition 8 (Bochner Spaces). Let $f : \Omega \rightarrow X$ be a given Banach-valued function.

1. We say $f \in L^p(\Omega; X)$ (called strongly measurable) if the quantity

$$\begin{cases} \int_{\Omega} \|f(y)\|_X^p dy, & \text{if } 1 \leq p < \infty \\ \text{ess sup}_{y \in \Omega} \|f(y)\|_X, & \text{if } p = \infty \end{cases} \quad (1.1)$$

is finite. The norm on $L^p(\Omega; X)$ is defined as

$$\|f\|_{L^p(\Omega; X)} = \begin{cases} \left(\int_{\Omega} \|f(y)\|_X^p dy \right)^{1/p}, & \text{if } 1 \leq p < \infty \\ \text{ess sup}_{y \in \Omega} \|f(y)\|_X, & \text{if } p = \infty \end{cases} \quad (1.2)$$

2. We say $f \in L_{weak}^p(\Omega; X)$ is weakly measurable, if
 - (a) the map $y \mapsto \|f(y)\|_X$ is measurable.
 - (b) quantities defined in 1.1 are finite. The norm is defined same as in 1.2.
3. We say $f \in L_{weak-}^p(\Omega; X^*)$ is weakly-* measurable, if
 - (a) the map $y \mapsto \|f(y)\|_{X^*}$ is measurable.
 - (b) The quantity

$$\begin{cases} \int_{\Omega} \|f(y)\|_{X^*}^p dy, & \text{if } 1 \leq p < \infty \\ \text{ess sup}_{y \in \Omega} \|f(y)\|_{X^*}, & \text{if } p = \infty \end{cases} \quad (1.3)$$

is finite. The norm on $L_{weak-}^p(\Omega; X^*)$ is defined as:

$$\|f\|_{L_{weak-}^p(\Omega; X^*)} = \begin{cases} \left(\int_{\Omega} \|f(y)\|_{X^*}^p dy \right)^{1/p}, & \text{if } 1 \leq p < \infty \\ \text{ess sup}_{y \in \Omega} \|f(y)\|_{X^*}, & \text{if } p = \infty \end{cases} \quad (1.4)$$

The characterization of dual spaces of $L^p(\Omega; X)$ when X is separable is as follows:

Theorem 3 (Duality of $L^p(\Omega; X)$). *Let X be a separable Banach space and let $1 \leq p < \infty$ with $1/p + 1/q = 1$. Then,*

$$[L^p(\Omega; X)]^* = L^q_{weak-*}(\Omega; X^*),$$

with the duality action defined as

$$\langle g, f \rangle_{L^q_{weak-*}(\Omega; X^*), L^p(\Omega; X)} = \int_{\Omega} \langle g(y), f(y) \rangle_{X^*, X} dy,$$

for any $g \in L^q_{weak-}(\Omega; X^*)$ and $f \in L^p(\Omega; X)$. If X is reflexive, then*

$$[L^p(\Omega; X)]^* = L^q(\Omega; X^*).$$

1.3 Young Measures

One of the key applications of Young measure in nonlinear analysis is the capturing of oscillations of weakly convergent sequences in L^1 .

A Young measure is a family of probability measures $\nu = \{\nu_y\}_{y \in \Omega} \in L^\infty_{weak-*}(\Omega; \mathcal{M}(\mathbb{R}^k))$ associated to a sequence $f_n : \Omega \rightarrow \mathbb{R}^k$ such that $\text{supp}(\nu_y) \subset \mathbb{R}^k$ and they depend measurably for $y \in \Omega$, which means that for any bounded continuous function $\phi : \mathbb{R}^k \rightarrow \mathbb{R}$, the function

$$\bar{\phi}(y) = \int_{\mathbb{R}^k} \phi(\lambda) d\nu_y(\lambda) =: \langle \nu_y, \phi \rangle \quad (1.5)$$

is measurable for every $y \in \Omega$.

Moreover, if we consider the composite sequence $\phi(f_n) : \Omega \rightarrow \mathbb{R}$ and assume that the sequence $\{\phi(f_n)\}_{n \in \mathbb{N}}$ converges weakly in $L^1(\Omega)$, then weak limit will precisely be the function given by 1.5.

Remark 3. Basically for any bounded, continuous function ϕ , the action $\langle \nu_y, \phi \rangle$ is measurable for every $y \in \Omega$.

Definition 9 (Carathéodary Function). A map $\psi : \Omega \times \mathbb{R}^k \rightarrow [-\infty, \infty]$ is a Carathéodary function if it is measurable in the first variable and continuous in the other variable i.e.

1. The map $\psi_{\lambda_0}(y) = \psi(y, \lambda_0)$ is measurable for each $\lambda_0 \in \mathbb{R}^k$.
2. The map $\psi_{y_0}(\lambda) = \psi(y_0, \lambda)$ is continuous for each $y_0 \in \Omega$.